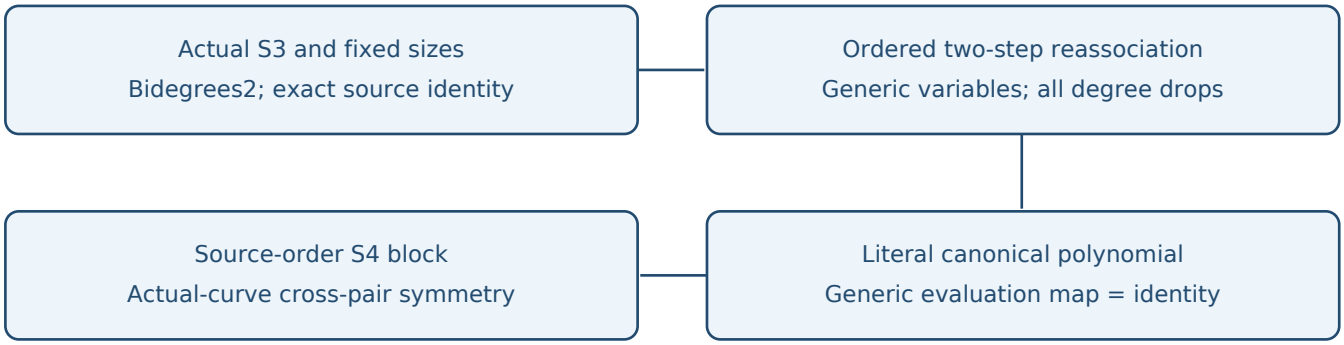
Actual S3 recursion: source-order reassociation

Proved component. Two actual recursive S3 resultant steps equal the source-ordered balanced expression with the existing S4 polynomial. Equality holds in the formal polynomial ring at every canonical stage. Signed standing: unverified. [pcc:unverified]

For deg P≤D, put $A(T)=\mathrm{Res}_U(P(U),S3(z,T,U);\ D,2)$. All sizes after the semicolon are declared Sylvester sizes and remain fixed under specialization. [pcc:unverified]

$\mathrm{Res}_T(A(T),S3(x,y,T);\ 2D,2)$
$= \mathrm{Res}_U(P(U),S4(z,x,y,U);\ D,4)$. [pcc:unverified]

The source-order theorem uses the actual b-invariants of a Weierstrass curve over a commutative integral domain. It includes zero P, repeated coordinates and every characteristic; nonsingularity is not needed for this polynomial identity. Actual recursivePolynomial W(n+2) uses D=2^(n+1). [pcc:unverified]



Proof. Literal S3 instantiates the fixed-size interchange theorem. Independent x/y variables followed by coefficient evaluation cover x=y. An independent leading coefficient specialized to0 covers every retained D≥deg P. The generic evaluation homomorphism is proved the identity on the ring of three independent variables, giving literal polynomial equality. Actual-curve S4 middle transposition and cyclic reordering then match the source argument order. [pcc:unverified]

Negative control. Arbitrary independent b-coefficients do not have the required S4 symmetry. For (b2,b4,b6,b8)=(0,0,0,1) over $\mathbb{Q}$, exchanging the middle two inputs of (0,1,2,3) changes the exact resultant. These coefficients violate the Weierstrass b-relation; this is not a qualified source counterexample. [pcc:unverified]

Scope. The canonical split with an S4 block (k=2, arity≥5) is normalized. General splits k≥3, higher full permutation symmetry and higher absolute irreducibility remain open. The all-fourteen audit retains its other original obligations. No new native experiment or performance claim. [pcc:unverified]

Evidence. Final whole Lean gate 60/0failures/0trusted; initial51gate retained separately. Lean4.35.0 / Mathlib56a00585…. Source/attempt/gate pins, exact declaration and proof map are in sources.json and evidence/. Standard foundations only; no custom axioms, native evaluation, raised limits or waivers. [pcc:unverified]

Sources. Semaev2004 Theorem1/p2; Kosters–Yeo2015 arXiv1503.08001v3 Section2/p3 general S3/recursion, text and PDF formula inspected; actual canonical/S4/evaluation source modules, Mathlib resultant maps and b-invariants. KB unavailable; bounded fallback, no novelty claim. Paid compute and agent uploads:0. [pcc:unverified]